

Jackson Houck

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EDUCATION

Boston College - Morrissey College of Arts and Sciences

Chestnut Hill, MA

Bachelor of Arts in Mathematics, Bachelor of Science in Computer Science

Expected May 2028

- **GPA:** 3.76/4.00; **ACT:** 35
- **Honors:** Dean's List (Every Semester)
- **Relevant Coursework:** Machine Learning, Natural Language Processing, Probability, Mathematical Statistics, Linear Algebra, Differential Equations, Multivariable Calculus, Data Structures, Foundations of Blockchain

EXPERIENCE

Research Assistant

Chestnut Hill, MA

Prud'hommeaux Lab @ Boston College

Sep. 2026 – Present

- Investigating adaptation of large language models to fifteen low-resource Formosan languages for native speakers

NSF REU Researcher - Natural Language Processing

Milwaukee, WI

University of Wisconsin-Milwaukee

Jun. 2026 – Aug. 2026

- Extended nanoGPT's PyTorch implementation of GPT-2 using token-aligned linguistic metadata at the embedding layer, reducing validation bits-per-byte by 3.6% against a compute-matched baseline on encyclopedic corpora
- Added a multi-task annotation-prediction head trained with binary cross-entropy alongside the language-modeling task, enabling the model to predict structural metadata at inference without requiring external annotations
- Wrote an algorithm for parsing nested markup, aligning subword annotations, and loading dual-stream memmaps; scaled training to multi-GPU SLURM jobs on Purdue Anvil through an ACCESS-CI allocation

Teaching Assistant

Chestnut Hill, MA

Boston College Department of Computer Science

Aug. 2025 – Jan. 2026; Aug. 2026 – Present

- Selected to TA Economics and Computation (Fall 2026), an upper-level Computer Science elective covering algorithmic game theory, auction design, mechanism design, matching, and cryptoeconomics
- Led weekly discussion sections and office hours for Computer Science 101 (Fall 2025), helping beginner students develop Python programming, debugging, and problem-solving skills; graded 40+ assignments weekly and provided actionable feedback on coding style, logical structure, implementation, and code readability

Executive Board Member

Chestnut Hill, MA

Heights Capital Corporation

Aug. 2024 – Present

- Served as an Executive Board analyst in the Macro, Quantitative, and Fixed Income Pod, delivering weekly market updates, analyzing economic releases, monetary-policy developments, and major market movements
- Led over 30 candidate interviews, organized alumni networking chats, and coordinated social events

PROJECTS

HER2 Breast Cancer Image Classification | *Python, PyTorch*

Mar. 2026 – May 2026

- Developed a ConvNeXt-Tiny system for four-class HER2 scoring from immunohistochemistry images, supporting patch-level training, whole-slide-style inference, and spatial model interpretation
- Designed a learned heatmap head with top- k pooling and a custom loss combining DAB-stain alignment, sparsity, and border penalties to localize relevant tissue while suppressing shortcut features

Options Volatility Surface Modeling | *Python, scikit-learn, PyTorch*

Feb. 2026 – Apr. 2026

- Engineered a reproducible options-data pipeline that transformed raw SPY and QQQ contracts into model-ready features including log-moneyness, time to expiration, volatility-index levels, and Treasury yields
- Compared random forest, gradient-boosting, multilayer-perceptron, and TabNet regressors, evaluating contract-level errors and reconstructing predicted volatility surfaces across strikes and maturities

Political Ideology Prediction | *Python, scikit-learn, spaCy, PyTorch*

Mar. 2026 – Apr. 2026

- Built an NLP classifier for 10,818 executive orders, comparing bag-of-words, TF-IDF, Word2Vec, CNN, and BERT-based approaches for predicting presidential party from formal political text
- Developed a spaCy entity-masking pipeline to remove names, dates, organizations, and political identifiers; achieved 83% accuracy and 0.84 F1 with a tuned CNN, substantially outperforming a 56.3% accuracy baseline

TECHNICAL SKILLS

Languages & Packages: Proficient: Python, NumPy; Familiar: Java, C++, C, scikit-learn, pandas, PyTorch

Activities: Eagle Scout, BC Blockchain (Vice President), Intramural Hockey & Soccer

Interests: Electric Guitar, Science Fiction, Homelabbing, European History, Warhammer 40k, Formula 1, BC Hockey